

We also know from before that: $\frac{\partial T}{\partial x} = \frac{\partial T_m}{\partial x} = \frac{2q''}{\rho C_p r_0 \bar{u}} = \text{constant}$

Back substituting $u(r)$ into our PDE

$$2\bar{u} \left[1 - \frac{r^2}{r_0^2} \right] \frac{\partial T}{\partial x} = \frac{\alpha}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \Rightarrow \text{PDE since } T(x, r)$$

Back substituting $\frac{\partial T}{\partial x}$:

$$\left(1 - \frac{r^2}{r_0^2} \right) \frac{4q''\bar{u}}{\rho C_p r_0 \bar{u}} = \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \Rightarrow \text{ODE} \Rightarrow \text{Solvable}$$

$$\boxed{\left(1 - \frac{r^2}{r_0^2} \right) \frac{4q''}{k r_0} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right)} \quad (1)$$

Integrating once, we obtain

$$r \frac{\partial T}{\partial r} = \frac{4q''}{k r_0} \left[\frac{r^2}{2} - \frac{r^4}{4r_0^2} \right] + C_1$$

$$\frac{\partial T}{\partial r} = \frac{4q''}{4r_0} \left[\frac{r}{2} - \frac{r^3}{4r_0^2} \right] + \frac{C_1}{r} \quad \leftarrow C_1 = 0 \text{ since } \frac{C_1}{r} \rightarrow \infty \text{ as } r \rightarrow 0$$

Integrating once more:

$$T(r) = \frac{4q''}{k r_0} \left[\frac{r^2}{4} - \frac{r^4}{16r_0^2} \right] + C_2$$

Using our second B.C.: $T(r=r_0) = T_0 \Rightarrow C_2 = T_0 - \frac{4q''}{k r_0} \left[\frac{3}{16} r_0^2 \right]$
So our solution becomes:

$$\boxed{T(r) = T_0 - \frac{4q''}{k r_0} \left(\frac{3r_0^2}{16} - \frac{r^2}{4} + \frac{r^4}{16r_0^2} \right)} \quad (2) \Rightarrow \text{Quartic temperature profile for } q'' = \text{const.}$$

Solving for T_m :

$$T_m = \frac{\int_0^{r_0} u T 2\pi r dr}{\pi r_0^2 \bar{u}} \Rightarrow \boxed{T_m = T_0 - \frac{11}{24} \cdot \frac{q'' r_0}{k}}$$

Note, $(T_m - T_0) = -\frac{11}{24} \cdot \frac{q'' r_0}{k} = \text{constant} \Rightarrow \text{makes sense according to our formulation.}$

Solving for Nusselt number:

$$h = \frac{q''}{T_o - T_m} = \frac{24k}{11r_o} = \frac{48k}{11D}$$

$$Nu_o = \frac{hD}{k} = \frac{48}{11} = 4.364$$

$$\boxed{Nu_o = 4.364} \Rightarrow \text{Laminar flow in tube with constant heat flux.}$$

Note, initially we solved that $Nu_o \sim 2$. Pretty close right!

Uniform Wall Temperature ($T_o = \text{constant}$)

For constant wall temperature, the solution proves difficult to solve analytically. Typically numerical solutions are obtained.

$q(x) = h[T_o - T_m] \Rightarrow T_m = f(x), T_o = \text{constant}$
we know from before that:

$$\frac{2T_m}{\partial x} = \frac{2q''}{r_o \rho c_p \bar{u}} \Rightarrow q'' = \frac{r_o \rho c_p \bar{u}}{2} \frac{\partial T_m}{\partial x} = h[T_o - T_m]$$

$$\text{let } \theta = T_o - T_m \Rightarrow d\theta = -dT_m$$

$$-\frac{r_o \rho c_p \bar{u}}{2} \frac{\partial \theta}{\partial x} = h\theta \Rightarrow \text{We can solve this}$$

$$\int_{\theta_1}^{\theta} \frac{\partial \theta}{\theta} = - \int_{x_1}^x \frac{2h}{r_o \rho c_p \bar{u}} dx$$

$$\ln \left(\frac{T_o - T_m}{T_o - T_{m,1}} \right) = - \frac{2h}{r_o \rho c_p \bar{u}} (x - x_1)$$

$$T_o - T_m = (T_o - T_{m,1}) \exp \left(- \frac{2h}{r_o \rho c_p \bar{u}} (x - x_1) \right) \Rightarrow \frac{1}{\rho c_p} = \frac{\alpha}{k}$$

$$\boxed{T_o - T_m = (T_o - T_{m,1}) \exp \left(- \frac{\alpha Nu}{r_o^2 \bar{u}} (x - x_1) \right)} \quad (1)$$

↳ The temperature difference decreases exponentially in the flow direction. As does the heat flux, q''

Now, we can say that: $T = T_0 - \phi(T_0 - T_m)$ scaling term.

$$\frac{\partial T}{\partial x} = \frac{\partial}{\partial x} [T_0 - \phi(T_0 - T_m)] = -\frac{\partial \phi}{\partial x} (T_0 - T_m) + \phi \frac{\partial T_m}{\partial x}$$

But since we have a fully developed flow; $\frac{\partial \phi}{\partial x} = 0$

$$\frac{\partial T}{\partial x} = \phi \frac{\partial T_m}{\partial x}$$

Back substituting everything into our energy equation:

$$\boxed{-2Nu(1-r^2)\phi = \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r}} \quad (2)$$

→ Similar to our $q'' = \text{constant}$ case but now ϕ term appears on the LHS.

We also need to use both boundary conditions to solve this numerically

$$\frac{\partial \phi}{\partial r} = 0 \text{ at } r=0 \quad (3)$$

$$\phi = 0 \text{ at } r=r_0 \quad (4)$$

Our last condition needed is our definition of Nu , since $\phi = f(r, Nu)$ in this problem:

$$Nu = \frac{hD}{k} = \frac{q''}{\Delta T} \cdot \frac{D}{k} = \frac{-k \left. \frac{\partial T}{\partial r} \right|_{r_0}}{\Delta T} \cdot \frac{D}{k} = -\left. \frac{\partial T}{\partial r} \right|_{r_0} \cdot D \cdot \frac{1}{T_0 - T_m}$$

We know $T = T_0 - \phi(T_0 - T_m)$

$$\frac{\partial T}{\partial r} = -\frac{\partial \phi}{\partial r} (T_0 - T_m) + \phi \frac{\partial}{\partial r} (T_0 - T_m)$$

$$Nu = \cancel{\left(-\frac{\partial \phi}{\partial r} \right) (T_0 - T_m)} \cdot D \cdot \frac{1}{\cancel{(T_0 - T_m)}} \quad \text{where } 0 \Rightarrow T_0, T_m \neq f(r)$$

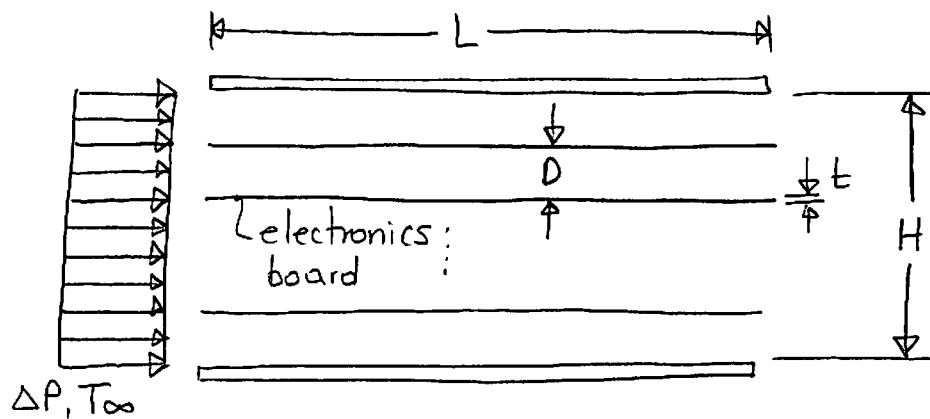
$$\boxed{Nu = 2r_0 \cdot \frac{\partial \phi}{\partial r}} \quad (5)$$

To solve (2), we need (3), (4) & (5). Solving numerically, we obtain:

$$\boxed{Nu_0 = \frac{hD}{k} = 3.66} \Rightarrow \text{Laminar flow in tube with constant wall temperature.} \quad (136)$$

Stack of Heat Generating Plates (Electronics cooling example)

Think of a stack (or rack) of electronics circuit boards



Incoming cooling fluid has a constant T_∞ , and pressure drop budget from a supply fan or pump.

Assuming the flow is laminar, and board temperature $T_w = \text{constant}$ ($<$ the electronics component breakdown temperature).
Also: $t \ll D$.

$$n = \frac{H}{D} \quad (\text{assuming } t \ll D)$$

Small Spacing limit: $D \rightarrow 0$

For $D \rightarrow 0$, we can say each channel becomes fully developed rapidly and remains fully developed for all L . Also, the fluid outlet temperature approaches T_w .

$$\bar{U} = \frac{D^2}{12\mu} \cdot \frac{\Delta P}{L} \Rightarrow \text{pg. (23) of notes} \Rightarrow \text{fully developed channel flow.}$$

$$\dot{m}' = \rho \bar{U} H = \rho H \frac{D^2}{12\mu} \cdot \frac{\Delta P}{L} \quad (\dot{m}' = \text{mass flow rate per unit depth})$$

For total heat transfer, since $T = T_w$ at $x = L$

$$q'_a = \dot{m}' c_p (T_w - T_\infty) = \rho H \frac{D^2}{12\mu} \cdot \frac{\Delta P}{L} \cdot c_p (T_w - T_\infty)$$

So $q'_a \sim D^2$ for the small spacing limit.